Surveillance Coverage, Treatment Precedence, and Workload Constraints in Machine Learning Early Warning for Sepsis: An Operational Evaluation

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Abstract

Objective: To evaluate how surveillance protocol coverage, treatment precedence, and emitted-alert constraints govern machine learning early warning for sepsis in an intensive care unit (ICU) cohort.

Methods: In a retrospective evaluation with sequential replay using 8,254 adult first ICU stays (length of stay $\geq$ 30h) from MIMIC-IV (primary) and MIMIC-III (secondary), 8 predictive architectures were evaluated under a prespecified second-day surveillance protocol (hours 19–30; 97,700 hourly windows; prevalence $\pi = 0.694\%$). We formalized an operational cascade (surveillance coverage, pretreatment opportunity, and alert-budget detection), evaluated pre-treatment risk sets excluding post-antimicrobial windows ($\mathcal{R}_0$ and $\mathcal{R}_2$), and estimated burden-matched operating frontiers under a 12-hour alert suppression rule.

Results: Among 408 consensus Sepsis-3 stays (4.94%), 257 (63.0%) had clinical onset after hour 36, precluding evaluation under the second-day protocol ($V = 0$). For 150 of 151 covered stays (99.3%), decision windows preceded antimicrobial administration. Model discrimination remained stable on pretreatment risk sets (BiMamba SSM: AUROC 0.6433 on $\mathcal{R}_0$ vs. 0.6326 canonical). Under an alert budget of 10 episodes per 100 patient-days, episode PPV ranged from 0.98% to 1.72%, corresponding to 58.1 to 101.8 simulated evaluations per detected patient (NNE), with whole-cohort timely yield bounded between 0.98% and 1.72%.

Discussion: Retrospective discrimination alone does not characterize clinical early-warning performance. In this restricted cohort, low empirical detection yield was jointly determined by surveillance coverage limits, modest baseline discrimination, and bedside alert-rate constraints, rather than discriminative collapse.

Conclusion: Sepsis early-warning systems must report surveillance coverage, verify treatment precedence, and evaluate operating characteristics against explicit alert-budget scenarios.

Keywords: Sepsis Early Warning, Surveillance Truncation, Treatment Precedence, Alert Budget Frontier, State-Space Models, Refractory Alert Suppression, Clinical Decision Support.

1 Introduction

1.1 Background and Significance

Sepsis remains a leading cause of critical illness, inpatient mortality, and healthcare expenditure globally [1, 2]. Clinical treatment urgency depends critically on presentation: the Surviving Sepsis Campaign international guidelines recommend immediate antimicrobial therapy, ideally within one hour, for possible septic shock or a high likelihood of sepsis, while recommending rapid diagnostic assessment for possible sepsis without shock [3, 4]. To support prompt intervention, the biomedical informatics community has heavily invested in machine learning (ML) models for early warning. Scores of retrospective investigations have reported high discriminative capability, frequently reporting area under the receiver operating characteristic curve (AUROC) values between 0.80 and 0.88 ahead of consensus onset [5]. However, published ICU prediction studies differ substantially in outcome implementation, temporal alignment, and evaluation design; consequently, reported AUROCs should not be interpreted as directly comparable measures of deployment readiness [5, 6].

Despite promising retrospective benchmarks, prospective hospital deployments have frequently yielded disappointing results [7, 10]. External evaluation of the original Epic Sepsis Model demonstrated that reported performance depended heavily on the evaluation level and timing rules: hospitalization-level and prediction-level workload estimates were not interchangeable, emphasizing the need to report the exact unit of evaluation [7]. Prospective evaluations of early-warning systems such as TREWS have found associations between timely alert confirmation and improved outcomes among selected patients identified prior to antibiotic initiation [8, 9]. However, because provider response was observational rather than randomized, these findings support potential clinical utility without establishing universal causal efficacy [8].

Furthermore, ICU clinicians already encounter substantial alarm exposure from bedside physiological monitors [13]. While bedside monitor alarms differ from algorithmic sepsis early-warning alerts, human-factors research demonstrates that repeated, low-information notifications induce cognitive fatigue and override behavior [14, 15].

This disconnect between retrospective benchmarks and prospective clinical utility is commonly attributed to discriminative collapse: the conjecture that ML models fail because they cannot anticipate sepsis prior to clinician recognition or empiric treatment [10]. Pretreatment evaluation has previously been proposed to distinguish genuine advance warning

from recognition of care processes already underway [11]. However, existing evaluations frequently conflate three distinct operational mechanisms:

1. **Surveillance Protocol Coverage (V):** Protocol surveillance windows (e.g., fixed observation periods) may miss episodes occurring outside the evaluated interval.
2. **Treatment Precedence (O):** Decision windows may occur after clinicians have already recognized infection and initiated antimicrobial therapy.
3. **Clinical Alert-Budget Constraints (D):** Even when models maintain pre-treatment discrimination, bedside alert-rate constraints bound real-world sensitivity and positive predictive value (PPV).

To isolate these mechanisms, we conducted an operational evaluation across 8 representative predictive architectures using audit-grade critical care streams from the Beth Israel Deaconess Medical Center (BIDMC) electronic health record (EHR) databases (MIMIC-IV v2.2 and MIMIC-III v1.4) [18, 19]. Our contributions are fourfold:

1. **Pretreatment Discrimination Benchmark:** We show that model discrimination does not collapse when evaluating true pre-treatment Sepsis-3 detection on risk sets excluding post-treatment windows ($\mathcal{R}_0$ and $\mathcal{R}_2$).
2. **Operational Alert-Budget Frontier:** We estimate retrospective burden-matched operating characteristics under a 12-hour alert suppression rule across realistic clinical workloads (5, 10, and 20 emitted alert episodes per 100 patient-days).
3. **Three-Stage Population Decomposition:** We formalize the cascade $S \to V \to O \to D$, demonstrating that under the specified schedule, 63.0% of sepsis episodes develop onset outside the surveillance window, whereas 99.3% of covered patients remain pre-treatment eligible.
4. **Multi-Clock Timestamp Uncertainty Analysis:** We bound prescription-proxy opportunity using calendar-day intervals in MIMIC-III, conditional on assigned Sepsis-3 timestamps and the observed surveillance schedule, and contrast them with confirmed electronic medication administration records (eMAR) in MIMIC-IV.

2 Methods

2.1 Study Design and Cohort Definition

We conducted a retrospective evaluation with sequential prediction replay using the MIMIC-IV critical care database (v2.2, primary evaluation) and the MIMIC-III database (v1.4, prescription-proxy sensitivity analysis) from BIDMC [18, 19]. The study cohort was defined as adult patients ($\geq$ 18 years) admitted to an intensive care unit with a length of stay LOS $\geq$ 30h. To avoid within-patient clustering across multiple admissions, only the chronological first ICU stay per patient was retained.

Surveillance was evaluated under a prespecified second-day ICU surveillance protocol: predictions were generated at 12 scheduled hourly decision bins $[t-1, t]$ for $t \in \{19, 20, \ldots, 30\}$ hours post-admission. The test partition comprised 8,254 unique patient ICU stays (97,700 evaluated windows, 4,070.8 patient-days of surveillance volume). Stays were censored at Sepsis-3 onset ($N = 110$ stays, 623 windows) or upon routine observation cessation under the prespecified feature-generation schedule ($N = 180$ stays, 725 windows). Reporting was organized in accordance with TRIPOD+AI and STROBE recommendations [20, 21].

2.2 Outcome Definition and Operational Clocks

Sepsis was defined according to consensus Sepsis-3 criteria [1, 2]: acute increase in Sequential Organ Failure Assessment (SOFA) score ≥ 2 within a clinical window $[-48\text{h}, +24\text{h}]$ of suspected infection (defined as blood culture specimen acquisition paired with systemic antimicrobial administration).

For each patient i, we tracked four distinct temporal clocks (Figure 1):

- $T_{\text{admit},i}$: ICU admission timestamp ($t = 0$).
- $T_{\text{treatment},i}$: First systemic intravenous antimicrobial administration.
- $T_{\text{culture},i}$: First blood culture specimen collection timestamp.
- $T_{\text{S3},i}$: Retrospective consensus Sepsis-3 onset timestamp.

2.3 Risk Sets, Anticipation Horizons, and Pre-treatment Eligibility

Let C denote the evaluated test cohort of unique ICU stays ($N = 8,254$), and let $\mathcal{T}_i \subseteq \{19, 20, \ldots, 30\}$ denote the set of hourly decision bins evaluated for stay i. For patients who never develop consensus Sepsis-3 during hospitalization, we set $T_{\text{S3},i} = \infty$.

The full evaluated early-warning risk set is defined as:

$$\mathcal{R} = \left\{(i,t) : i \in C,\ t \in \mathcal{T}_i,\ t < T_{\text{S3},i}\right\} \tag{1}$$

The binary early-warning prediction target at anticipation horizon $H = 6$h is:

$$Y_i(t; H) = \mathbf{1}\left\{t < T_{\text{S3},i} \leq t + H\right\} \tag{2}$$

In the canonical unconstrained risk set ($\mathcal{R}_{\text{canonical}} = \mathcal{R}$), this definition yields exactly 97,700 evaluated hourly windows, comprising 678 horizon-positive observations ($Y = 1$) and 97,022 negative observations ($Y = 0$, prevalence $\pi = 0.694\%$).

To evaluate true pre-treatment detection without feature distortion from prior antimicrobial exposure, we defined the pre-treatment risk sets $\mathcal{R}_\delta$ for clinical buffer $\delta \geq 0$:

$$\mathcal{R}_\delta = \left\{(i,t) \in \mathcal{R} : t + \delta \leq T_{\text{treatment},i}\right\} \tag{3}$$

where $T_{\text{treatment},i} = \infty$ for stays with verified absence of antimicrobial therapy during hospitalization. Windows occurring after treatment initiation ($t + \delta > T_{\text{treatment},i}$) were excluded from both cases and controls. We evaluated $\mathcal{R}_0$ ($\delta = 0$h, censoring windows ending after treatment) and $\mathcal{R}_2$ ($\delta = 2$h, requiring a 2-hour pre-treatment actionability buffer).

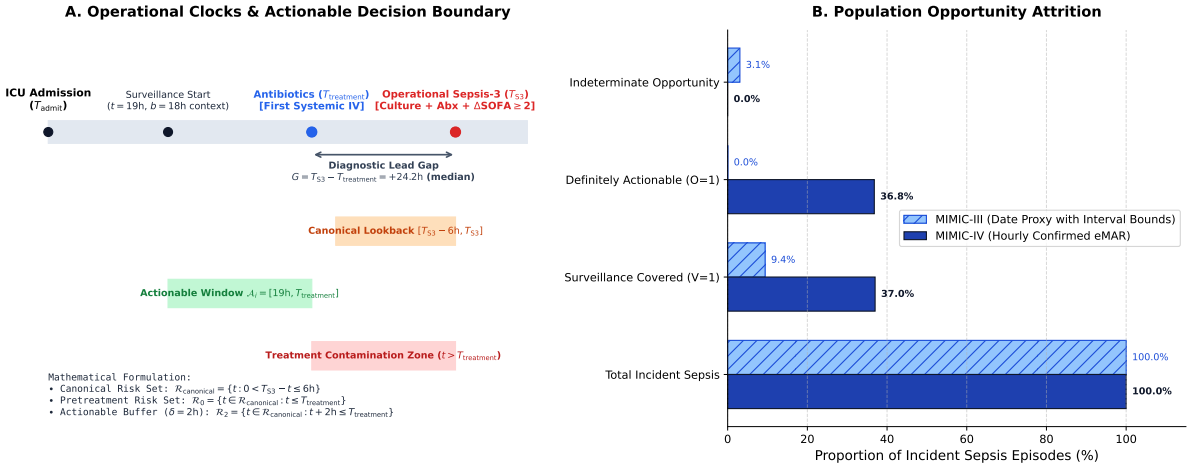


Figure 1: Operational Clocks, Pretreatment Decision Windows, and Population Opportunity Attrition. Panel A illustrates the clinical timeline for an ICU-incident sepsis patient, showing the diagnostic lead gap $G = T_{S3} - T_{\text{treatment}} = +24.2$h (median), the actionable decision window $\mathcal{A}_i$, and the post-treatment observation period. Panel B displays population opportunity attrition comparing MIMIC-III (with calendar-day interval-censored prescription bounds) and MIMIC-IV (hourly confirmed eMAR administration records).

2.4 Alert-Budget Operating Frontiers and Refractory Policy

To quantify simulated emitted-alert burden, we evaluated fixed alert budgets $q \in \{5.0, 10.0, 20.0\}$ emitted alert episodes per 100 surveillance patient-days. Under a 12-hour suppression rule, each stay emits at most one alert episode within the evaluated 12-hour interval ($N_{\text{episodes},i} \leq 1$).

We estimated retrospective burden-matched operating characteristics by selecting score cutoffs from the held-out score distribution. These operating points describe empirical performance at matched emitted-alert rates; they do not represent thresholds prospectively fixed for online deployment. Patient selection used the maximum score over the evaluated interval, whereas detection timing was assessed at the first eligible threshold crossing under the complete tie-selection rule:

$$t_{\text{first},i} = \min\{t \in \mathcal{T}_i : \text{Elig}_i(t; \tau, \mathcal{B}) = 1\} \tag{4}$$

where $\mathcal{B}$ denotes the deterministic prioritization rule:

$$\text{Elig}_i(t; \tau, \mathcal{B}) = \mathbf{1}\{i \in \mathcal{S}_{\text{alerted}}(K) \ \wedge \ s_i(t) \geq \tau\} \tag{5}$$

with $\mathcal{S}_{\text{alerted}}(K)$ defined as the top K patient replicas under the lexicographical sort:

$$\left(-\max_t s_i(t),\ \text{ICUSTAY_ID},\ \text{copy_idx}\right)$$

2.5 Predictive Models and Statistical Inference

We evaluated 8 models spanning three major architectural paradigms:

1. **State-Space Models:** Bidirectional Mamba (BiMamba SSM) utilizing linear-time selective state-space layers [16, 17].

2. **Deep Sequence Transformers:** Mask-Aware Clinical Transformer (incorporating explicit missingness masks) and Baseline Unmasked Transformer.

3. **Tabular and Ensemble Baselines:** Random Forest, XGBoost, LightGBM, Logistic Regression (L2), and a Multi-Model Stacking Ensemble.

All models were trained on fixed pre-surveillance development partitions with hyperparameters and feature normalization frozen prior to this operational evaluation.

Uncertainty was quantified using patient-clustered paired bootstrapping ($B = 1,000$ resamples). For Procedure B (Empirical Burden-Matched Bootstrap), exposure $E_b = \frac{1}{24}\sum_{i \in b} h_i$ and alert target $K_b = \text{round}(q \cdot E_b/100)$ were dynamically recomputed within each resample. NNE confidence intervals were derived by exact reciprocal transformation of unrounded PPV percentiles: $\text{NNE}_{\text{low}} = 100/\text{PPV}_{\text{upper}}$ and $\text{NNE}_{\text{high}} = 100/\text{PPV}_{\text{low}}$ (with $+\infty$ when $\text{PPV}_{\text{low}} = 0$).

3 Results

3.1 Cohort Progression and Denominator Accounting

Table 1 traces all 8,254 ICU admissions through surveillance eligibility, onset timing, lookback horizons, and pre-treatment actionability. Among 408 consensus Sepsis-3 stays, 257 (63.0%) had clinical onset after hour 36 (median onset: 49.0h post-admission). Under the specified observation schedule and 6-hour horizon, these 257 stays had no evaluated horizon-positive decision time ($V = 0$). The remaining 151 stays (37.0%) had active protocol surveillance coverage ($V = 1$). Crucially, for 150 of these 151 covered patients (99.3%), decision windows occurred at or before recorded antimicrobial treatment, providing definitive pre-treatment opportunity ($\mathcal{R}_0$).

3.2 Pretreatment Discrimination Benchmark

Table 2 presents discrimination across Canonical Sepsis-3, Pretreatment Sepsis ($\mathcal{R}_0$), and Pretreatment with 2h Buffer ($\mathcal{R}_2$). Model discrimination remained stable or slightly higher in pre-treatment risk sets (BiMamba SSM: AUROC 0.6326

Table 1: Cohort Progression and Denominator Accounting Ledger. Denominators traced from the base test cohort through surveillance eligibility, sepsis onset timing, prediction horizons, and pre-treatment actionability.

Progression Stage / Cohort Partition	Stays (N)	Evaluated Windows	Surveillance Volume	Prevalence	Clinical Definition & Measurement Convention
1. Base Prospective-Replay Test Cohort	8,254	97,700 / 99,048 sched.	4,070.8 pt-days	100.0%	Adult first ICU stays ≥ 30h; 1,348 censored (623 onset, 725 cessation)
2. Non-Sepsis Control ICU Stays	7,846	93,427 neg. windows	3,892.8 pt-days	95.06%	Stays without consensus Sepsis-3 during hospitalization
3. Consensus Sepsis-3 Stays ($S = 1$)	408	4,273 (678 pos, 3,595 neg)	178.0 pt-days	4.94%	Blood culture + antibiotic + acute ΔSOFA ≥ 2; median onset 49.0h
3a. Late-Onset Sepsis (Onset > 36h)	257	3,084 neg. windows	128.5 pt-days	63.0% of septic	Onset > 36h; lookback $[T_{S3} - 6, T_{S3}]$ does not intersect [19, 30] ($V = 0$)
3b. Surveillance-Covered ($V = 1$)	151	1,189 (678 pos, 511 neg)	49.5 pt-days	37.0% of septic	Onset in [20h, 36h]; lookback intersects surveillance window ($V = 1$)
4. Canonical Sepsis-3 Risk Set (H = 6h)	8,254	97,700 (678 pos / 97,022 neg)	4,070.8 pt-days	0.694%	Full unconstrained risk set ($t \in \mathcal{T}_i,\ t < T_{S3,i}$)
5. Pretreatment Risk Set $\mathcal{R}_0$ (δ = 0h)	5,967	70,192 (674 pos / 69,518 neg)	2,924.7 pt-days	0.960%	Censoring post-treatment windows ($t \leq T_{\text{treatment}}$); 150/151 covered eligible
6. Pretreatment Risk Set $\mathcal{R}_2$ (δ = 2h)	5,967	70,129 (663 pos / 69,466 neg)	2,922.0 pt-days	0.945%	2-hour clinical buffer ($t + 2\text{h} \leq T_{\text{treatment}}$); 149/151 covered eligible

canonical vs. 0.6433 on $\mathcal{R}_0$, paired ΔAUROC = +0.0107, 95% CI: +0.0060 to +0.0175; Stacking Ensemble: 0.6284 vs. 0.6351, paired ΔAUROC = +0.0067). The increase in AUPRC (BiMamba: 1.209% to 1.656%) reflects higher baseline positive prevalence in the restricted set (0.694% vs. 0.960%). These results are directionally consistent with the antibiotic-order sensitivity findings of Dutta et al. (2026) [12], though differences in population selection, predictive models, and clinical-action timestamps preclude treating this as a direct replication.

3.3 Observed Burden–Detection Trade-Off Under This Protocol

Table 3 and Figure 2 demonstrate the real operational constraint: under realistic alert budgets, clinical detection yield is severely bounded.

At an emitted workload of 10 alert episodes per 100 patient-days (407 emitted episodes across 4,070.8 patient-days):

- Across all 8 models, episode-level PPV ranged from 0.98% to 1.72%, corresponding to 58.1 to 101.8 simulated evaluations per detected patient (NNE).
- BiMamba SSM and LightGBM each uniquely detected **7 patients** (PPV = 1.72%, 95% CI: 0.49%–2.95%; NNE = 58.1, 95% CI: 33.9–203.5).
- Stacking Ensemble, XGBoost, and Random Forest each detected **6 patients** (PPV = 1.47%, NNE = 67.8).
- All detected patients strictly preceded antimicrobial therapy by ≥ 2 hours ($t_{\text{first}} + 2\text{h} \leq T_{\text{treatment}}$).
- Model rankings varied across budgets: at 20 episodes/100 pt-days, XGBoost detected 15 patients (NNE = 54.3) and Mask-Aware Transformer detected 14 patients (NNE = 58.1), highlighting workload-dependent operating behavior.

3.4 Two-Axis Subgroup and Timestamp Uncertainty Matrix

Table 4 presents the cross-tabulation of Sepsis Onset Timing against Treatment Timing. In MIMIC-IV, eMAR administration timestamps verify that 150 of 151 covered patients (99.3%) and 36.8% of total incident sepsis episodes had definitive pre-treatment prediction opportunity. In MIMIC-III, prescription start dates lack sub-day precision, producing calendar-day intervals [date at midnight, next date at midnight) that generate 110 indeterminate cases across the cohort, bounding prescription-proxy opportunity between 0.03% and 3.10% conditional on assigned consensus Sepsis-3 onset times.

3.5 Cumulative Population Coverage Decomposition

Figure 3 illustrates the cumulative population coverage decomposition:

$$\frac{151}{408} \times \frac{150}{151} \times \frac{7}{150} = \frac{7}{408} = 1.72\%$$

1. **Surveillance Truncation** ($151/408 = 37.0\%$)**:** 63.0% of sepsis patients develop onset after hour 36, outside the prespecified second-day surveillance protocol.
2. **Pretreatment Actionability** ($150/151 = 99.3\%$)**:** Among covered stays, treatment almost never occurs prior to surveillance lookback windows.
3. **Workload Detection Yield** ($7/150 = 4.67\%$)**:** Under an alert budget of 10 episodes/100 patient-days, BiMamba SSM detects 7 patients (1.72% whole-population yield; NNE = 58.1).

4 Discussion

This investigation resolves the operational trade-offs governing machine learning early warning for sepsis in ICU surveillance:

1. **Pre-treatment Discrimination Is Preserved:** In directional consistency with Dutta et al. (2026), model discrimination (AUROC) does not collapse when evaluating true pre-treatment sepsis detection on risk sets excluding post-treatment windows ($\mathcal{R}_0$ and $\mathcal{R}_2$).
2. **Detection Yield Is Constrained by Workload and Prevalence:** AUROC values near 0.64 are modest. Under

Table 2: Benchmark Discrimination Across Pretreatment Tasks (Held-Out Point Estimates and Clustered 95% CIs). Point estimates calculated directly on held-out test data ($B = 0$); 95% CIs derived from $B = 1,000$ patient-clustered bootstrap resamples.

Model Architecture / Metric	Canonical Sepsis-3 (H = 6h)	Pretreatment $\mathcal{R}_0$ (δ = 0h)	Pretreatment $\mathcal{R}_2$ (δ = 2h)	Paired ΔAUROC ($\mathcal{R}_0$ − Canon)
BiMamba SSM (Proposed)	0.6326 (0.5872–0.6733)	**0.6433** (0.5998–0.6835)	**0.6447** (0.6002–0.6860)	**+0.0107** (+0.0060 to +0.0175)
AUPRC	*1.209% (0.892–1.769)*	***1.656%** (1.235–2.329)*	***1.647%** (1.220–2.331)*	—
Multi-Model Stacking Ensemble	0.6284 (0.5829–0.6699)	0.6351 (0.5897–0.6771)	0.6370 (0.5913–0.6799)	+0.0067 (+0.0021 to +0.0137)
AUPRC	*1.206% (0.882–1.748)*	*1.621% (1.207–2.263)*	*1.613% (1.198–2.255)*	—
Random Forest	0.6108 (0.5643–0.6535)	0.6143 (0.5684–0.6575)	0.6173 (0.5712–0.6612)	+0.0034 (-0.0008 to +0.0101)
AUPRC	*1.080% (0.821–1.434)*	*1.470% (1.138–1.920)*	*1.461% (1.120–1.916)*	—
Mask-Aware Clinical Transformer	0.5942 (0.5467–0.6410)	0.5961 (0.5490–0.6422)	0.5954 (0.5480–0.6426)	+0.0020 (-0.0024 to +0.0083)
AUPRC	*1.013% (0.766–1.303)*	*1.382% (1.048–1.760)*	*1.365% (1.032–1.752)*	—
Tabular GBDT (XGBoost)	0.5670 (0.5208–0.6102)	0.5643 (0.5186–0.6082)	0.5652 (0.5192–0.6099)	-0.0027 (-0.0070 to +0.0038)
AUPRC	*0.977% (0.739–1.357)*	*1.320% (1.008–1.817)*	*1.313% (0.998–1.811)*	—
Tabular GBDT (LightGBM)	0.5613 (0.5161–0.6026)	0.5587 (0.5139–0.6010)	0.5598 (0.5145–0.6033)	-0.0026 (-0.0070 to +0.0038)
AUPRC	*0.893% (0.694–1.195)*	*1.222% (0.957–1.629)*	*1.214% (0.949–1.630)*	—
Logistic Regression (L2)	0.5593 (0.5097–0.6048)	0.5654 (0.5157–0.6107)	0.5657 (0.5150–0.6127)	+0.0061 (+0.0023 to +0.0116)
AUPRC	*0.939% (0.716–1.218)*	*1.337% (1.012–1.740)*	*1.325% (1.002–1.731)*	—
Clinical Transformer (Baseline)	0.5239 (0.4808–0.5653)	0.5296 (0.4851–0.5712)	0.5286 (0.4834–0.5705)	+0.0057 (+0.0019 to +0.0110)
AUPRC	*0.808% (0.609–1.381)*	*1.147% (0.857–1.904)*	*1.131% (0.845–1.894)*	—

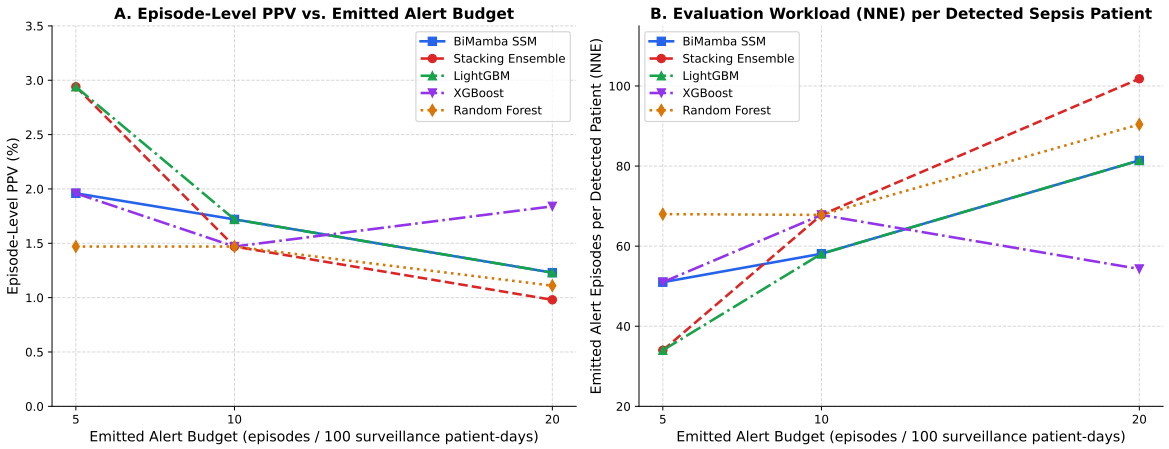


Figure 2: Operational Alert-Budget Performance Frontier. Panel A displays episode-level positive predictive value (PPV %) across prespecified alert budgets. Panel B displays the Number Needed to Evaluate (NNE) per detected patient under a 12-hour alert suppression rule. Lines illustrate trends across evaluated budget tiers; complete bootstrap confidence intervals and boundary-degenerate notes are reported in Table 3.

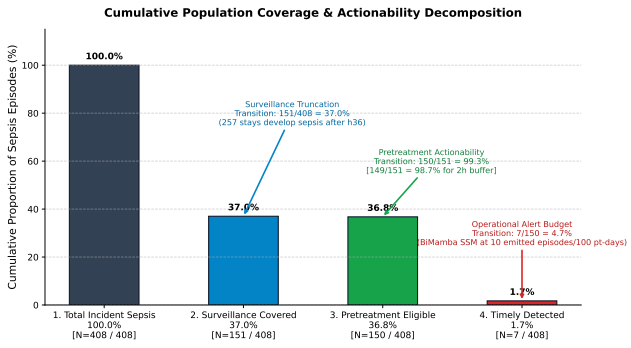


Figure 3: Cumulative Population Coverage and Actionability Decomposition. Stepwise attrition from 408 incident sepsis episodes through protocol coverage, pre-treatment opportunity, and alert-budget detection.

low clinical alert budgets (≤ 10 alert episodes per 100 patient-days), whole-population sensitivity is bounded below 2%, with PPV between 0.98% and 1.72%, requiring 58.1 to 101.8 simulated bedside reviews per detected patient (NNE).

3. **Surveillance Protocol Coverage Explains Unreached Cases:** Under the evaluated second-day protocol, 63.0% of incident cases develop onset after the monitoring window terminates, preventing multi-hour warning.

A critical conceptual error in prior literature is **endpoint substitution**. When benchmarks predict "antibiotic initiation within 6 hours across all patients", they evaluate care-process prediction (predicting clinician ordering behavior, including prophylactic prescribing in non-septic patients). This task is fundamentally distinct from pre-treatment sepsis detection. Machine learning benchmarks must maintain the operational Sepsis-3 outcome definition as the target while conditioning evaluation timestamps on decision availability.

Furthermore, repeated and low-information notifications have been associated with reduced clinician acceptance in other decision-support settings [14]. While we quantify simulated emitted-alert exposure and evaluation yield, our analysis did not measure cognitive overload, alert acceptance, or alarm fatigue directly [14, 15]. Clinical impact also depends on whether alerts are reviewed and acted upon [9]. Our evaluation estimates opportunities and simulated review burden; it does not estimate provider adoption or changes in treatment timing.

Table 3: Operational Burden-Matched Alert Frontiers Under 12h Refractory Suppression. Evaluated under Procedure B with exposure recomputation and first-emitted-alert timing verification.

Budget	Model Architecture	Cutoff (τ)	Alerts (K)	Episode PPV % [95% CI]	Det (N)	Sens. Cov %	Yield %	NNE [95% CI]
5.0 ep / 100 pt-d	Stacking Ensemble	0.0145	204	2.94 (0.98–5.88)%	6	3.97 (1.31–7.65)%	1.47 (0.49–2.86)%	34.0 (17.0–102.0)
	LightGBM	0.0117	204	2.94 (0.98–5.39)%	6	3.97 (1.26–7.10)%	1.47 (0.47–2.61)%	34.0 (18.5–102.0)
	BiMamba SSM	0.0187	204	1.96 (0.49–4.45)%	4	2.65 (0.65–6.20)%	0.98 (0.24–2.31)%	51.0 (22.5–204.0)
	XGBoost	0.0122	204	1.96 (0.49–3.94)%	4	2.65 (0.63–5.63)%	0.98 (0.23–2.09)%	51.0 (25.4–204.0)
	Random Forest	0.0177	204	1.47 (0.00–3.45)%	3	1.99 (0.00–4.49)%	0.74 (0.00–1.70)%	68.0 (29.0–∞)
	Clinical Transformer	0.0101	204	0.98 (0.00–2.46)%	2	1.32 (0.00–3.50)%	0.49 (0.00–1.27)%	102.0 (40.6–∞)
	Logistic Regression	0.0103	204	0.98 (0.00–2.94)%	2	1.32 (0.00–3.85)%	0.49 (0.00–1.45)%	102.0 (34.0–∞)
	Mask-Aware Transformer	0.0120	204	0.00 (0.00–0.00)%*	0	0.00 (0.00–0.00)%*	0.00 (0.00–0.00)%*	Undefined (∞)
10.0 ep / 100 pt-d	BiMamba SSM	0.0165	407	1.72 (0.49–2.95)%	7	4.64 (1.33–8.16)%	1.72 (0.49–2.93)%	58.1 (33.9–203.5)
	LightGBM	0.0110	407	1.72 (0.49–2.95)%	7	4.64 (1.38–7.74)%	1.72 (0.49–2.89)%	58.1 (33.9–203.5)
	Stacking Ensemble	0.0131	407	1.47 (0.25–2.46)%	6	3.97 (0.70–6.98)%	1.47 (0.26–2.54)%	67.8 (40.7–407.0)
	XGBoost	0.0114	407	1.47 (0.49–2.94)%	6	3.97 (1.32–7.65)%	1.47 (0.49–2.85)%	67.8 (34.0–203.5)
	Random Forest	0.0158	407	1.47 (0.49–2.70)%	6	3.97 (1.26–7.29)%	1.47 (0.47–2.70)%	67.8 (37.0–204.0)
	Mask-Aware Transformer	0.0110	407	1.23 (0.25–2.46)%	5	3.31 (0.74–6.67)%	1.23 (0.26–2.48)%	81.4 (40.7–407.0)
	Clinical Transformer	0.0094	407	1.23 (0.25–2.46)%	5	3.31 (0.68–6.38)%	1.23 (0.25–2.37)%	81.4 (40.7–407.0)
	Logistic Regression	0.0095	407	0.98 (0.25–1.97)%	4	2.65 (0.59–5.51)%	0.98 (0.23–2.07)%	101.8 (50.8–408.0)
20.0 ep / 100 pt-d	XGBoost	0.0105	814	1.84 (0.98–2.70)%	15	9.93 (5.26–14.47)%	3.68 (1.87–5.29)%	54.3 (37.0–101.9)
	Mask-Aware Transformer	0.0099	814	1.72 (0.74–2.58)%	14	9.27 (4.54–13.77)%	3.43 (1.59–5.19)%	58.1 (38.8–135.7)
	Logistic Regression	0.0088	814	1.35 (0.49–2.09)%	11	7.28 (2.74–11.11)%	2.70 (0.99–4.18)%	74.0 (47.8–203.7)
	BiMamba SSM	0.0142	814	1.23 (0.49–2.09)%	10	6.62 (3.07–10.89)%	2.45 (1.05–4.10)%	81.4 (47.9–203.2)
	LightGBM	0.0105	814	1.23 (0.61–2.09)%	10	6.62 (3.11–11.04)%	2.45 (1.15–4.10)%	81.4 (47.9–163.0)
	Random Forest	0.0131	814	1.11 (0.49–1.85)%	9	5.96 (2.70–9.68)%	2.21 (0.96–3.69)%	90.4 (54.2–203.7)
	Stacking Ensemble	0.0115	814	0.98 (0.37–1.72)%	8	5.30 (2.04–9.15)%	1.96 (0.74–3.43)%	101.8 (58.1–271.3)
	Clinical Transformer	0.0088	814	0.86 (0.25–1.48)%	7	4.64 (1.51–8.18)%	1.72 (0.52–3.04)%	116.3 (67.8–407.0)

*At zero observed detections, the empirical percentile-bootstrap interval was degenerate at zero. This does not establish zero population detection probability. NNE intervals derived by exact reciprocal transformation. 100% of detected patients satisfied pre-treatment timing buffers (δ = 0h and δ = 2h).

Table 4: Two-Axis Subgroup and Timestamp Uncertainty Matrix. Opportunity bounds conditional on fixed consensus Sepsis-3 timestamps and surveillance coverage.

Cohort & Timestamp Source	Sepsis Onset Category	Total (N)	Covered (V = 1)	Eligible (O = 1)	Indeterminate	Subgroup Bounds	Contribution to Cohort
MIMIC-IV Prospective-Replay	Early Incident (≤ 24h)	47	47 (100.0%)	47	0	[100.0%, 100.0%]	47 / 408 (11.5%)
(Hourly Confirmed eMAR)	Intermediate (24–48h)	156	104 (66.7%)	103	0	[66.0%, 66.0%]	103 / 408 (25.2%)
	Late Incident (> 48h)	205	0 (0.0%)	0	0	[0.0%, 0.0%]	0 / 408 (0.0%)
	All Incident Sepsis-3	**408**	**151 (37.0%)**	**150**	**0**	**[36.8%, 36.8%]**	**150 / 408 (36.8%)**
MIMIC-III Broad Incident Sepsis	Early Incident (≤ 24h)	2,641	102 (3.9%)	0	74	[0.0%, 2.8%]	[0, 74] / 3,581 (0.00–2.07%)
(Date-Level Prescription Proxy)	Intermediate (24–48h)	361	235 (65.1%)	1	36	[0.3%, 10.2%]	[1, 37] / 3,581 (0.03–1.03%)
	Late Incident (> 48h)	579	0 (0.0%)	0	0	[0.0%, 0.0%]	[0, 0] / 3,581 (0.00%)
	All Incident Sepsis-3	**3,581**	**337 (9.4%)**	**1**	**110**	**[0.03%, 3.10%]**	**[1, 111] / 3,581 (0.03–3.10%)**

4.1 Limitations

The analysis estimates operating characteristics in a selected long-stay ICU cohort rather than clinical effectiveness. Burden-matched cutoffs were derived retrospectively from complete evaluated score trajectories, and identifier-based selection was required for tied boundary scores. Accordingly, the frontier does not represent a prospectively locked policy or guarantee a specified alert rate in online deployment.

Observed treatment precedence does not establish that infection was unrecognized or that earlier antimicrobial administration would have been appropriate. The study did not measure clinician response, review duration, treatment changes, alarm fatigue, or patient outcomes; NNE therefore represents a simulated evaluation count rather than measured clinical workload. Alarm-reduction interventions may suppress both nonactionable and clinically important signals; our short evaluation interval cannot establish the safety or optimal duration of a refractory policy during sustained surveillance [15]. Finally, both MIMIC-III and MIMIC-IV originate from a single medical center (BIDMC); multi-center validation across diverse health systems remains essential.

5 Conclusion

In this restricted, long-stay ICU cohort, observed detection yield depended on surveillance coverage, baseline discrimination, and the policy used to constrain emitted alerts. Most horizon-positive observations remained pretreatment eligible, and model rankings varied across workload levels. Deployable sepsis early-warning informatics requires benchmarking under explicitly specified emitted-alert policies and retrospectively burden-matched operating points, deduplicating refractory alert workload, and enforcing multi-clock reporting.

Ethics Approval and Consent to Participate

This study was conducted using de-identified data from the MIMIC-III (v1.4) and MIMIC-IV (v2.2) databases, collected at Beth Israel Deaconess Medical Center (BIDMC). Data collection was approved by the Institutional Review Boards of BIDMC and the Massachusetts Institute of Technology (MIT). The data were de-identified in accordance with Health Insurance Portability and Accountability Act (HIPAA) Safe Harbor standards; informed consent was waived by the reviewing IRBs. Access was granted under PhysioNet Credentialed Data Use Agreement.

Competing Interests

The authors declare that they have no competing financial or non-financial interests.

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Authors' Contributions

AS conceived the study, developed the analytical framework, implemented the pipelines, performed the experiments, and drafted the manuscript. SA provided academic supervision, contributed to methodological design, critically revised the manuscript, and approved the final draft.

Code and Data Availability

All preprocessing pipelines, model configurations, and release manifests are publicly available under MIT License at `https://github.com/alireza/Sepsis-Onset-Timing-Paradox` (Git commit `b0932ca`, manifest `1.0.0-freeze`). De-identified patient data are available from PhysioNet under the MIMIC-IV Data Use Agreement.

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